Raw .tsg data vs NVCL API data

What each source contains, what the classifier uses, and what could improve it · GSWA HyLogger Industry Adoption (Group 6)

> **Scope:** every number in this document comes from **one hole, 05GJD001** (25,938 scan points, 150–316 m). The raw package was read with pyTSG; the API was queried live with nvcl_kit. Findings are not yet validated across other holes.

1. Short terms used in this document

Term	Meaning
NVCL	National Virtual Core Library. The open archive of HyLogger spectral scans and core photos; the WA holes are the ones GSWA published.
HyLogger	The instrument that scans drill core with infrared light and identifies minerals at each depth.
API / nvcl_kit	nvcl_kit is the Python client library that calls the NVCL web services. "API data" in this document means data fetched through it.
WFS	Web Feature Service. The standard geospatial web service nvcl_kit uses to list boreholes.
TSG / pyTSG	The Spectral Geologist (CSIRO software that produces the files). pyTSG is GSWA's Python package that reads a hole's TSG file package.
.tsg / .bip	.tsg is the index/header file. .bip holds the binary data (spectra and mineral results). NIR/SWIR and TIR each have their own pair.
.cal / .ini	Calibration and instrument/processing settings. Used automatically by pyTSG.
cras.bip / hires.dat	cras.bip is a high-resolution core line-scan image (373 MB). hires.dat is a core-surface profilometer log (height profile), not spectral.
NIR / SWIR / VNIR	Near-, short-wave- and visible-near-infrared light. SWIR (about 1000–2500 nm) sees OH-bearing minerals: clays, micas, chlorite, carbonate. VNIR (visible) sees iron oxides.
TIR	Thermal infrared (about 6000–14500 nm). Sees silicates such as quartz and feldspar that SWIR barely detects.
TSA	The Spectral Assistant. The automatic mineral-identification module inside TSG. It is a machine guess, not a geologist's log.
uTSAS, sTSAS, sTSAV	TSA result sets. The last letter is the wavelength range: S = SWIR, V = VNIR (visible), T = thermal. The first letter (u, s, d) is a processing variant; its exact meaning is not confirmed by us.
ujCLST, sjCLST	TIR classification result sets (the thermal counterpart of the TSA sets). Exact expansion not confirmed.
Grp1 / Min1	Grp1 = broad mineral group of the top match (e.g. CHLORITE). Min1 = the specific top mineral (e.g. Chlorite-FeMg). Grp2/3, Min2/3 are runner-up matches.
Wt1	Weight of the top match, 0 to 1. Used as a confidence signal.
Error, SNR	Error = fit uncertainty of the match. SNR = signal-to-noise ratio of the spectrum.
Scalar	One interpreted value per depth sample (e.g. Grp1 uTSAS). The pyTSG scalar table has 151 columns for the SWIR scan.
w2200 / w2250	Wavelength (nm) of the absorption feature near 2200 nm (white mica) and 2250 nm (chlorite). Continuous numbers, not category labels.
OXIDE	The sTSAV group for iron-oxide minerals. In this hole it is almost all goethite (4,929 of its 4,937 samples); the other 8 are hematite.
INVALID / NOTAROK	Flagged results that are not usable mineral calls. Excluded from mineral percentages.
Sentinel	The no-data marker −3.402823e38 stored in numeric columns (blank in text columns).
Native resolution	The instrument's own sampling, about 8 mm along the core (25,938 points in this hole).

Bin	A depth window (e.g. 20 m) into which native samples are grouped for percentages.
pp	Percentage points (difference between two percentages).
total_n	Number of valid classified samples in a bin. The classifier needs at least 20.
WAMEX / A-number	WA exploration report database. The A-number is a report ID; 05GJD001's is A73613, found in the .tsg header text.

2. What each source contains (hole 05GJD001)

Data item	NVCL API (nvcl_kit)	Raw files (pyTSG)	Notes
uTSAS Grp1 / Min1 / Wt1 / Error (SWIR)	Yes	Yes	All 20 published API logs (uTSAS and ujCLST: Grp, Min, Wt, Error) were compared with the raw file over all 25,938 rows. Text is identical; numeric differences are below 1e-12.
Native ~8 mm rows	Yes	Yes	The API returns every native row. The 20 m resolution is a setting in `etl/fetch_holes.py`.
TIR classes (ujCLST): quartz, feldspars, pyroxene, spodumene	Yes	Yes	Published as API scalar logs; labels verified identical to raw. Also has TIR spectra (raw only).
sTSAS (second SWIR method)	No	Yes	The API lists only 20 scalar logs for this hole: uTSAS and ujCLST.
sTSAV (visible: OXIDE)	No	Yes	Min1 is stored as numeric codes. Decoded with pyTSG's class table: Goethite 4,929; Aspectral 4,399; GalvanisedIron 136; ODZincalume 8; Hematite 8.
Named indices (w2200, w2250, ferric oxide abundance, hematite-goethite distribution)	No	Yes	Continuous per-sample values, 6,500–14,500 valid samples each.
Water scalars, SNR, Flags, absorption features (25 per sample)	No	Yes	Part of the 151 SWIR scalar columns.
Raw spectra (NIR 531 bands, TIR 341 bands)	Spectral log empty	Yes	Needed only if we build our own model on spectra.
Profilometer (hires.dat)	List empty	Yes	Core surface height profile. Not yet inspected further.
WAMEX report ID, GSWA UWI, map sheet, archive location	Not checked	Yes	In the .tsg header text; pyTSG's structured API does not expose it.
Tray photos and mosaic	Yes	Yes	33 trays plus one stitched mosaic.

3. What the implemented classifier does

File: `classifier/rules_baseline.py`. Inputs per depth interval: `total_n`, WHITE-MICA, CHLORITE, HEMATITE, KAOLIN, GOETHITE (percentages). Rules run in order and the first match wins. All thresholds are team placeholders; there is no ground truth yet.

Rule	Condition	Label	Can it fire on current data?

R4	total_n < 20	Unclassified	**Possible, but did not fire in any tested window (smallest total_n was 28 at 1 m).**
R1	White mica ≥ 40% and chlorite ≥ 25%	Altered zone	**Rarely. Fired once in 166 one-metre windows.**
R2	Hematite ≥ 15% and chlorite ≥ 15%	Possible ore zone	**Effectively never. Hematite is 0 samples in SWIR and only 8 samples in the whole hole (sTSAV, raw file only).**
R3	Kaolin + goethite ≥ 55% and white mica < 15%	Waste rock	**Not from SWIR: goethite is absent there and kaolin is only 39 samples. Goethite exists only in sTSAV (raw file), and the result then depends on how its percentage is normalised (finding 3).**
R5	No rule matched	Unclassified	**Fires for almost everything.**

4. Findings and their impact on the classifier

#	Finding	Evidence (05GJD001)	Why it matters
1	Ore and waste rules cannot fire	SWIR Grp1 has 0 HEMATITE and 0 GOETHITE samples. Result: 9 of 9 bins Unclassified at 20 m, 34 of 34 at 5 m, 165 of 166 at 1 m.	The classifier cannot output its main labels from uTSAS alone.
2	Goethite and hematite exist only in sTSAV	sTSAV Min1 decoded with pyTSG's class table: Goethite 4,929 samples, Hematite 8, Aspectral 4,399, GalvanisedIron 136, ODZincalume 8. Not published by the API for this hole.	Goethite (needed by rule R3) requires the raw file. Hematite (rule R2) is almost absent from the whole hole.
3	Real sTSAV goethite and hematite: R2 still cannot fire; R3 depends on normalisation	Hematite is at most 1.0% of any 20 m bin, so R2 never fires. R3 (waste) with goethite as % of all samples: 0 of 9 bins (20 m), 0 of 34 (5 m), 4 of 166 (1 m). As % of valid VNIR calls: 8 of 9, 28 of 34, 127 of 166.	Swings from almost never to almost always depending on one normalisation choice, so the rule needs a client-defined basis before it is trusted. Chlorite-rich intervals are being labelled waste in the second case.
4	The pipeline pulls too little from the API	fetch_holes.py requests only Grp1 uTSAS at 20 m. Wt1, Error and TIR are already published.	Free improvement: confidence and TIR without any raw download.
5	The 20 m sampled pull does not match native data	With INVALID excluded on both sides, native uTSAS rebinned to the same 20 m bins differs from the API pull by a mean 4.9 pp per mineral per bin, over bin–mineral pairs present in either dataset (largest: DARK-MICA 11.8, CHLORITE 11.3, CARBONATE 7.5). Bin tallies differ too (valid samples: 841 vs 1,703 at 150–170 m; 2,045 vs 1,525 at 210–230 m). The pull returned 8 of the 10 mineral groups (no SMECTITE, EPIDOTE).	Cause unconfirmed (the endpoint's bin edges are not documented to us). Native rows avoid the problem.
6	Confidence varies a lot between samples	Among 14,252 valid uTSAS mineral calls: mean Wt1 0.67, 13.5% below 0.5, 37.6% below 0.6, 12.1% exactly 1.0. All 1,754 INVALID samples also sit at Wt1 = 1.0 (half of the 3,473 samples at 1.0).	The classifier treats every sample as equally certain. INVALID must be removed before using Wt1. The remaining 1,719 valid calls at exactly 1.0 are unexplained.
7	TIR covers more of the hole than SWIR	TIR classified 18,456 samples (18,441 valid) vs uTSAS 16,006 (14,252 valid). 2,520 samples have a TIR mineral only. Quartz (SILICA) 12,497.	Quartz, feldspar and spodumene are invisible to the classifier today.

8	Two SWIR methods mostly agree	uTSAS vs sTSAS, sample by sample: of 14,247 samples where both give a mineral, 96.6% agree (3.4% disagree). 703 samples have a mineral from only one of the two. sTSAV is a different mineral family and is not compared.	Agreement can serve as a per-interval confidence flag.
9	Coverage differs by method	Share of 25,938 samples with any result / with a valid mineral call: uTSAS 61.7% / 54.9%, sTSAS 63.3% / 57.6%, sTSAV 36.5% / 19.0%.	About 45% of the hole has no valid uTSAS mineral call (38.3% no result plus 6.8% INVALID).

5. Possible next steps

#	Action	Source needed	Expected benefit
1	Change the ETL to pull native rows plus Wt1, Error and TIR (ujCLST) from the API	API only	Confidence weighting, TIR minerals, no 20 m tally gap.
2	Extract sTSAV goethite / hematite using pyTSG's class table and agree how to normalise them against the SWIR percentages	Raw file + client	Makes rule R3 testable with real goethite; shows whether R2 has any data at all.
3	Add a cross-method agreement flag (uTSAS vs sTSAS)	Raw file	Per-interval confidence beyond Wt1.
4	Try continuous indices (w2200, w2250, ferric oxide) as features	Raw file	Less dependence on brittle percentage thresholds.
5	Ask GSWA whether Wt1 = 1.0 is genuine and what the 20 m sampled endpoint does	Client	Resolves findings 5 and 6.
6	Get client-confirmed ore, waste and alteration definitions	Client	Replaces placeholder thresholds; enables real evaluation.
7	Repeat the comparison on 3–5 more holes	API + raw	Checks that these findings are not specific to 05GJD001.